



Focus Laser Scanning Solution

**Capture With
Confidence
and Connect
Your World
Faster**

Hybrid Reality
Capture™,
powered by Flash
Technology™



Performance Specifications

Models	Focus Premium Max	Focus Premium	Focus Core
Unambiguity Interval ¹	614 m		
Range			
White, 90% Reflectivity	0.5 – 400 m	0.5 – 200 m	0.5 – 100 m
Dark-grey, 10% Reflectivity	0.5 – 150 m	0.5 – 150 m	0.5 – 100 m
Black, 2% Reflectivity	0.5 – 50 m	0.5 – 50 m	0.5 – 50 m
Range Noise ^{2,3}			
White, 90% Reflectivity	0.1 mm @ 10 m, 0.2 mm @ 25 m		
Dark-grey, 10% Reflectivity	0.3 mm @ 10 m, 0.4 mm @ 25 m		
Black, 2% Reflectivity	0.7 mm @ 10 m, 1.2 mm @ 25 m		
Max Speed	Up to 2 MPts/sec		
3D Accuracy ⁴	2 mm @ 10 m, 3.5 mm @ 25 m		
Ranging Error ⁵	±1 mm		
Angular Accuracy ⁶	19 arcsec		
LaserHDR	Yes		
Temperature Range ⁷	Operating: +5° to +40° C, Extended Operating: -10° to +55° C, Storage: -10° to +60° C		

Additional Performance Specifications

Color Unit	
Color Resolution	Up to 266 MPx color
Raw Color Resolution	867 MPx Core: 527 MPx
HDR Camera	13 MPx - 2x, 3x, 5x brackets Core: 8 MPx - 2x, 3x, 5x brackets
Parallax	Minimized due to co-axial design
Deflection Unit	
Field of View	300° vertical ⁸ / 360° horizontal
Step Size	0.009° (40,960 Pts on 360°) vertical / 0.009° (40,960 Pts on 360°) horizontal
Max. Scan Speed	97 Hz (vertical)
Laser (Optical Transmitter)	
Laser Class	Laser Class 1
Wavelength	1553.5 nm
Beam Divergence	0.3 mrad (1/e)
Beam Diameter at Exit	2.12 mm (1/e)
Data Handling and Control	
Data Storage	SATA 3.0 SSD 128 GB and SDXC™ V30 64 GB SD Card; SD3.0, UHS-I / SDXC™ / SDHC™, max. 512 GB
Scanner Control	Via touch screen display and WLAN connection, Control by FARO Stream App (iOS & Android) or mobile devices with HTML5
Interface Connection	
WLAN	IEEE 802.11 ac/a/b/g/n 2x2 MIMO, as access point or client in existing networks (2.4 and 5 GHz)
USB	USB 3 port

Additional Features

Dual Axis Compensator	Performs a leveling of each scan with an accuracy of 19 arcsec valid within ±2°
Height Sensor	Via an electronic barometer, the height relative to a fixed point can be detected and added to a scan
Compass ⁹	The electronic compass gives the scan an orientation
GNSS	Integrated GPS & GLONASS
On-Site Compensation	Creates current quality report and improves compensation automatically
Accessory Bay	The accessory bay connects versatile accessories to the scanner
Inverse Mounting	Yes
Real-time, On-site Registration	Stream App real-time scan streaming, registration, overview map and Sphere cloud upload
Electronic Automation Interface	Available as option, only at point of sale
Digital Hash Function	Scans are cryptographically hashed and signed by the scanner
Rescanning of Distant Targets	Defined areas recaptured in higher resolution at a greater distance
Retake Photos	Select individual photographs with unwanted objects and retake them

General Specifications

Power Supply	19 V (external supply), 14.4 V (internal battery)
Typical Power Consumption	19 W idle, 32 W scanning, 72 W charging
Typical Battery Operation Time	About 4 hours
Typical Scan Time: From start until the scanner can be moved ¹⁰	Gray scale < 1 min HDR Colored < 1:15 Colored Flash Scan < 30 seconds ¹¹
Ingress Protection (IP) Rating Class	54
Humidity	Non-condensing
Weight	4.4 kg (including battery)
Size/Dimensions	230 x 183 x 103 mm
Calibration	Recommended annually
Manufacturer Warranty	2 year



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